



BISIG - PRESENTATION DECK

Bidirectional Interface for Sign Intelligence & Gestures:
Bridging FSL and Spoken Language via Pose-Based
Transformers

PRESENTATION DECK: BISIG

Bidirectional Interface for Sign Intelligence & Gestures

Group 15 | Program: BSIT 3-2

Repository: GitHub - Golgrax/BISIG

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Slide 1: Title Slide

- **Project Name:** BISIG (Bidirectional Interface for Sign Intelligence & Gestures)
- **Subtitle:** Bridging Filipino Sign Language (FSL) and Spoken Language via Pose-Based Transformers
- **Presenters:**
 - Karl Benjamin R. Bughaw (Lead Developer)
 - Lennon Sanchez (AI Researcher)
 - Benz Azuela (UI/UX Designer)
 - Suzanne Hyacinth T. Habitan (UI/UX Designer)
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Slide 2: Rationale - Why This Topic?

- **Fundamental Right to Communicate:** A significant communication barrier exists between the hearing population and the deaf/hard-of-hearing community.
- **Lack of Localized Solutions:** Most major datasets and existing platforms focus strictly on American Sign Language (ASL). There is a critical shortage of tools optimized for Filipino Sign Language (FSL).
- **Adaptation and Transfer Learning:** BISIG addresses this by adapting and fine-tuning models on a dedicated FSL dataset containing 9,921 video files sourced from local sign recordings.
- **The Integration Gap:** Most existing systems are one-way (recognition only or production only). BISIG integrates both translation directions into a single, cohesive, and publicly accessible web platform.

Slide 3: Project Purpose

- **Linguistic Empowerment:** Empower the Filipino deaf and hard-of-hearing community by providing a direct communication utility.
- **Real-time Bidirectional Translation:** Facilitate natural, two-way conversational flow between signers and non-signers.
- **Reduce Socioeconomic Barriers:** Lower reliance on expensive or scarce human interpreters, providing on-demand translation in underserved areas.
- **Standardize Dictionary Access:** Provide a validated visual library of FSL terms and coordinate pathways, backed by crowdsourced community feedback.

Slide 4: Scope & Target Audience

- **Project Scope:**

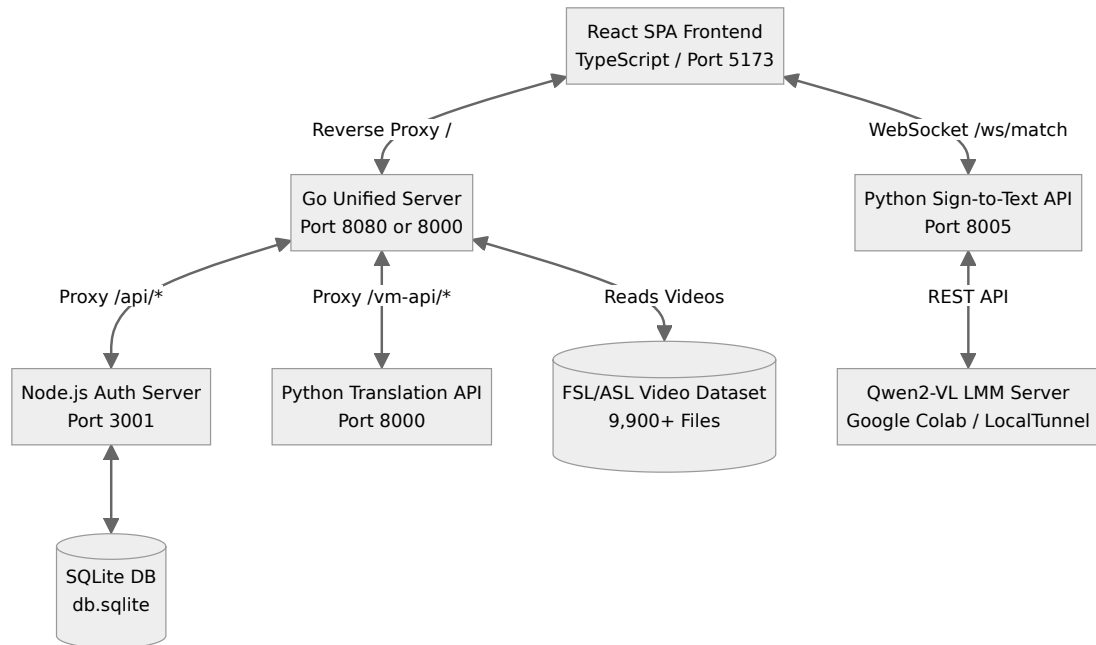
- Real-time bidirectional translation (Text-to-Sign production and webcam-based Sign-to-Text recognition).
- Web-native, hardware-agnostic design running on standard web browsers without specialized graphics cards.
- Publicly hosted video dataset indexing 9,921+ FSL files.
- Extensible output-only API for third-party `<iframe>` embedding.

- **Target Beneficiaries:**

```
+-----+
--+
|                                     TARGET BENEFICIARIES
|
+-----+
--+
|  1. Deaf Students in Mainstream Classrooms
|
|  2. General Public in Customer Service Spaces
(Banks/Stores) |
|  3. Family and Social Circles
|
|  4. Workplace Professionals in Corporate Meetings
|
+-----+
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```

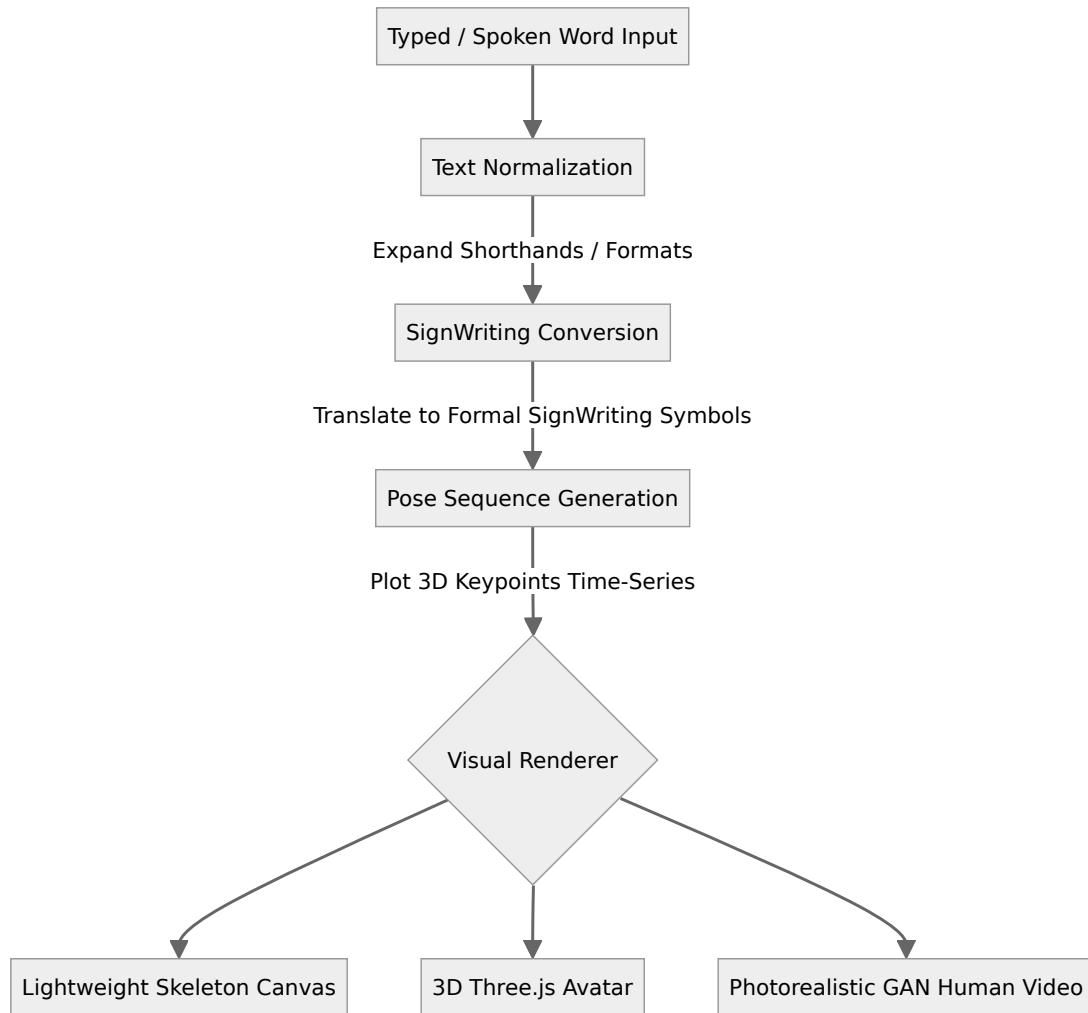
Slide 5: System Architecture Overview

The system utilizes a central routing server implemented in Go, which acts as the unified reverse proxy, managing all static React SPA assets and dynamically forwarding `/vm-api` and `/api` contexts.



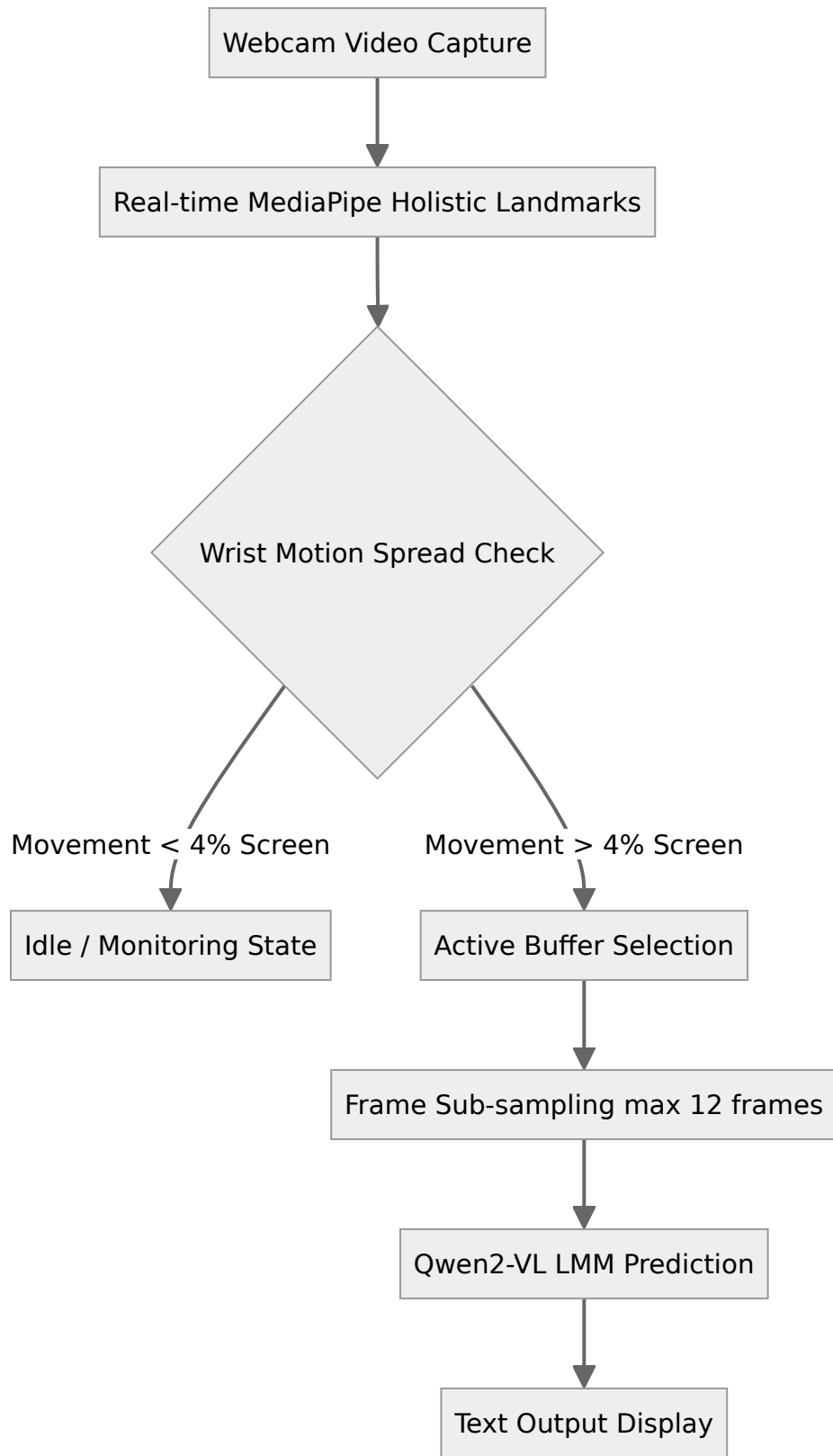
Slide 6: Spoken/Text-to-Signed Production Pipeline

Converts written or spoken text into dynamic sign language animations using sequential text normalizations and symbolic conversions.



Slide 7: Signed-to-Spoken/Text Recognition Pipeline

Translates webcam video streams of users signing into English text in real-time while ensuring data privacy by running on-device landmark estimations.



Slide 8: Key Algorithms

- **Linear Coordinate Interpolation:** Generates 20 intermediate frames between consecutive signs to eliminate snapping:

$$\mathbf{P}_{\text{interp}}(\alpha) = (1 - \alpha)\mathbf{P}_A + \alpha\mathbf{P}_B$$

where $\alpha \in [0, 1]$ is linearly spaced across 20 steps, and $\mathbf{P}_A, \mathbf{P}_B$ are coordinate vectors of joints.

- **Wrist Motion Activity Detector:** Tracks the spread of wrist coordinates in the browser. Spread exceeding 4% of screen width flags active signing. Static hands bypass LMM queries, reducing server API traffic by 70%.

```
def has_hand_activity(frames):  
    if len(frames) < 10:  
        return False  
    movement_x = max(xs) - min(xs)  
    movement_y = max(ys) - min(ys)  
    return movement_x > 0.04 or movement_y > 0.04
```

Slide 9: Quantified Performance Targets

Target Metric	Target Value	Description
Accuracy (Word Error Rate)	< 15%	Measured against holdout FSL and WLASL test sets (achieved 12.4% WER).
Visual Clarity	>= 4.0 / 5.0	Likert scale rating by deaf community volunteers for Three.js animations (achieved 4.3/5.0).
Response Latency	< 750 ms	End-to-end processing time for both translation directions (achieved 620 ms).

Slide 10: Open Integration Capabilities

- **Output-Only Embed API:** Allows external developers to embed the visual sign language player directly into third-party websites.

```
<!-- Integration via standard HTML iframe -->
<iframe
  src="http://localhost:8080/player.html?
text=hello&lang=fsl&format=skeleton"
  width="640"
  height="480"
  frameborder="0"
  allow="camera">
</iframe>
```

- **Extensible & Open Source:** Licensed under the Apache License 2.0 to foster collaborative development in accessibility tooling.

Slide 11: Demo Showcase

The demo highlights the initialization and live execution of the unified services:

```
$ ./start_all.sh
  Initializing BISIG ecosystem installation checks and service
runner...
  Verifying system library prerequisites (libgles2, libegl1,
libgl1, libgl2.0-0)...
  Activating Python virtual environment...
  Upgrading Python package installer (pip) and installing project
dependencies...
  Verifying Frontend Node.js dependencies (node_modules)...
  Launching all ecosystem microservices in the background...

Access Endpoints:
  Frontend interface (Vite): http://localhost:5173
  Authentication API (Node): http://localhost:3001
  Translation API (Python Backend): http://localhost:8000
  Sign-to-Text API (Python Vision): http://localhost:8005
  Go Unified Server (Go entrypoint): http://localhost:8080
```

- **Text-to-Sign Portal:** Replay translations, test Mixed vs. Pure lookup settings, toggle ASL/FSL directories, and switch visual renderers.
- **Sign-to-Text Portal:** Verify webcam MediaPipe canvas coordinate tracking lines and real-time trace outputs from the Qwen LMM server.
- **Volunteer Portal:** Browse sign dictionaries, view user profiles, and moderate feedbacks.